

3.4. Geometric applications.

We now use the tools we have seen in a range of applications.

Example: Let $\vec{u} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} -1 \\ 3 \\ -1 \end{bmatrix}$, and $\vec{w} = \begin{bmatrix} 2 \\ -1 \\ -4 \end{bmatrix}$.

(i) Are $\vec{u}, \vec{v}, \vec{w}$ linearly independent?

(ii) Are $\vec{u}, \vec{v}, \text{proj}_{\vec{v}} \vec{u}$ linearly independent? Justify it using a volume.

(iii) Are $\vec{u}, \vec{v}, \text{proj}_{\vec{w}} \vec{v}$ linearly independent? Justify it without computing $\text{proj}_{\vec{w}} \vec{v}$.

(i) We are solving the equation:

$$s_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + s_2 \begin{bmatrix} -1 \\ 3 \\ -1 \end{bmatrix} + s_3 \begin{bmatrix} 2 \\ -1 \\ -4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

which gives the system of equations:

$$\begin{cases} s_1 - s_2 + 2s_3 = 0 \\ 2s_1 + 3s_2 - s_3 = 0 \\ 3s_1 - s_2 - 4s_3 = 0 \end{cases}$$

which can be solved via the following Gaussian elimination.

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 2 & 3 & -1 & 0 \\ 3 & -1 & -4 & 0 \end{array} \right] \xrightarrow[\substack{R_2 - 2R_1 \\ R_3 - 3R_1}]{} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 0 & 5 & -5 & 0 \\ 0 & 2 & -10 & 0 \end{array} \right] \xrightarrow[\substack{\frac{1}{5} \cdot R_2 \\ \frac{1}{2} \cdot R_3}]{}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & -5 & 0 \end{array} \right] \xrightarrow{R_3 - R_1} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & -6 & 0 \end{array} \right] \xrightarrow{\frac{1}{-6} \cdot R_3}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

This augmented matrix in row echelon form corresponds to the system:

$$\begin{cases} s_1 - s_2 + 2s_3 = 0 \\ s_2 - s_3 = 0 \\ s_3 = 0 \end{cases}$$

which has the unique solution $s_1 = s_2 = s_3 = 0$. The vectors $\vec{u}$, $\vec{v}$, $\vec{w}$ are linearly independent.

(ii) We first compute:

$$\text{proj}_{\vec{v}} \vec{u} = (\vec{u} \cdot \vec{v}) \cdot \frac{\vec{v}}{\|\vec{v}\|^2} = \left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 3 \\ -1 \end{bmatrix} \right) \cdot \frac{1}{1^2 + 3^2 + 1^2} \cdot \begin{bmatrix} -1 \\ 3 \\ -1 \end{bmatrix} = \frac{2}{11} \cdot \begin{bmatrix} -1 \\ 3 \\ -1 \end{bmatrix} = \begin{bmatrix} -2/11 \\ 6/11 \\ -2/11 \end{bmatrix}$$

The volume of the parallelepiped determined by $\vec{u}$, $\vec{v}$, and $\text{proj}_{\vec{v}} \vec{u}$ is:

$$\left| \det \begin{bmatrix} 1 & 2 & 3 \\ -1 & 3 & -1 \\ -2/11 & 6/11 & -2/11 \end{bmatrix} \right| = 1 \cdot (-6/11 + 6/11) + 2 \cdot (2/11 - 2/11) + 3 \cdot (-6/11 + 6/11) = 0.$$

This means that $\vec{u}$, $\vec{v}$, and $\text{proj}_{\vec{v}} \vec{u}$ are in the same plane, so they are linearly dependent.

(iii) The vectors $\vec{v}$ and $\vec{w}$ are not orthogonal because:

$$\vec{v} \cdot \vec{w} = (-1) \cdot 2 + 3 \cdot (-1) + (-1) \cdot (-4) = -1 \neq 0.$$

Thus $\text{proj}_{\vec{w}} \vec{v} \neq \vec{0}$. Since $\text{proj}_{\vec{w}} \vec{v}$ is not zero and points in the same

direction as $\vec{w}$, then the three vectors $\vec{u}, \vec{v}, \text{proj}_{\vec{w}} \vec{v}$ are linearly

independent if and only if the three vectors $\vec{u}, \vec{v}, \vec{w}$ are linearly

independent. We saw in (i) that $\vec{u}, \vec{v}, \vec{w}$ are linearly independent, so

$\vec{u}, \vec{v}, \text{proj}_{\vec{w}} \vec{v}$ are linearly independent.

Example: Find all values k for which the plane $x_1 + \frac{3}{5}x_2 + (k-1)x_3 = 1$ and the

plane $\vec{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \frac{s}{3} \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix} + t \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$ intersect, and a parametric form of the intersection.

We first find an equation form of the plane given in parametric form. A vector

orthogonal to both $\begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$ is their crossed product.

$$\begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix} \times \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} = \det \begin{bmatrix} \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \\ 0 & 2 & 3 \\ -1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} -5 \\ -3 \\ 2 \end{bmatrix}$$

So the second plane has an equation form:

$$0 = (\vec{x} - \vec{\frac{1}{3}}) \cdot \vec{v} = \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right) \cdot \begin{bmatrix} -5 \\ -3 \\ 2 \end{bmatrix} = -5x_1 - 3x_2 + 2x_3 + 6, \text{ namely}$$

$$-5x_1 - 3x_2 + 2x_3 = -6.$$

We now solve the system of equations given by the two planes.

$$\begin{cases} x_1 + \frac{3}{5}x_2 + (k-1)x_3 = 1 \\ -5x_1 - 3x_2 + 2x_3 = -6 \end{cases}$$

$$\left[\begin{array}{ccc|c} 1 & \frac{3}{5} & k-1 & 1 \\ -5 & -3 & 2 & -6 \end{array} \right] \xrightarrow{R_2 + 5 \cdot R_1} \left[\begin{array}{ccc|c} 1 & \frac{3}{5} & k-1 & 1 \\ 0 & 0 & 5k-3 & -1 \end{array} \right]$$

When $5k-3=0$, the system has no solutions. When $5k-3 \neq 0$ the system will

have infinitely many solutions. Thus when $k = \frac{3}{5}$ the planes do not intersect.

When $k \neq \frac{3}{5}$ we can continue simplifying the augmented matrix.

$$\left[\begin{array}{ccc|c} 1 & \frac{3}{5} & k-1 & 1 \\ 0 & 0 & 5k-3 & -1 \end{array} \right] \xrightarrow{\frac{1}{5k-3} \cdot R_2} \left[\begin{array}{ccc|c} 1 & \frac{3}{5} & k-1 & 1 \\ 0 & 0 & 1 & \frac{-1}{5k-3} \end{array} \right]$$

This row echelon form has exactly one free parameter, so the planes

intersect at a line. We can find the line explicitly by writing the system

$$\begin{cases} x_1 + \frac{3}{5}x_2 + (k-1)x_3 = 1 \\ x_3 = \frac{-1}{5k-3} \end{cases}$$

which after substituting the second equation into the first becomes

$$\begin{cases} x_1 + \frac{3}{5}x_2 = \frac{4k-2}{5k-3} \\ x_3 = \frac{-1}{5k-3} \end{cases}$$

To write a parametric form corresponding to it, we set $x_2 = s$ a parameter.

Then we have:

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{4k-2}{5k-3} - \frac{3}{5}s \\ s \\ \frac{-1}{5k-3} \end{bmatrix} = \begin{bmatrix} \frac{4k-2}{5k-3} \\ 0 \\ \frac{-1}{5k-3} \end{bmatrix} + s \begin{bmatrix} -\frac{3}{5} \\ 1 \\ 0 \end{bmatrix}.$$

Example: Problems of the form "find the next number in the sequence" make no

sense. We can cook up polynomials that produce any number that we want.

A sequence of numbers begins as: 1, 2, 3, 4. Find a polynomial satisfying

$f(1)=1$, $f(2)=2$, $f(3)=3$, $f(4)=4$, and $f(5)=k$ where k is a fixed scalar.

We first need to determine the degree of the polynomial $f(x)$. Since we

have five equalities that need to be satisfied, we need a polynomial with

five parameters (which will be the unknowns we will be solving for). We

need $f(x) = ax^4 + bx^3 + cx^2 + dx + e$, a polynomial of degree four. Then:

$$\begin{cases} 1 = f(1) = a + b + c + d + e \\ 2 = f(2) = 16a + 8b + 4c + 2d + e \\ 3 = f(3) = 81a + 27b + 9c + 3d + e \\ 4 = f(4) = 256a + 64b + 16c + 4d + e \\ k = f(5) = 625a + 125b + 25c + 5d + e \end{cases}$$

is the system of equations we are solving, using Gaussian elimination.

$$\left[\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 1 \\ 16 & 8 & 4 & 2 & 1 & 2 \\ 81 & 27 & 9 & 3 & 1 & 3 \\ 256 & 64 & 16 & 4 & 1 & 4 \\ 625 & 125 & 25 & 5 & 1 & k \end{array} \right] \xrightarrow{\text{rref}} \left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & 0 & \frac{1}{24} \cdot k - \frac{5}{24} \\ 0 & 1 & 0 & 0 & 0 & -\frac{5}{12} \cdot k + \frac{25}{12} \\ 0 & 0 & 1 & 0 & 0 & \frac{35}{24} \cdot k - \frac{175}{24} \\ 0 & 0 & 0 & 1 & 0 & -\frac{25}{12} \cdot k + \frac{137}{12} \\ 0 & 0 & 0 & 0 & 1 & k - 5 \end{array} \right]$$

The polynomial we want is then:

$$f(x) = \left(\frac{1}{24} \cdot k - \frac{5}{24} \right) x^4 + \left(-\frac{5}{12} \cdot k + \frac{25}{12} \right) x^3 + \left(\frac{35}{24} \cdot k - \frac{175}{24} \right) x^2 + \left(-\frac{25}{12} \cdot k + \frac{137}{12} \right) x + k - 5.$$

Example: Find the intersection between the line $\vec{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + r \begin{bmatrix} 1 \\ 1 \\ 2 \\ 2 \end{bmatrix}$ and the plane

$$\vec{x} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \\ 3 \\ 2 \end{bmatrix}.$$

We are solving the following equation:

$$\begin{bmatrix} 1+r \\ r \\ 1+2r \\ 2r \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + r \begin{bmatrix} 1 \\ 1 \\ 2 \\ 2 \end{bmatrix} = \vec{x} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} s+t \\ 1-s+t \\ 1+s+3t \\ 2t \end{bmatrix}$$

namely the system of equations:

$$\begin{cases} r - s - t = -1 \\ r + s - t = 1 \\ 2r - s - 3t = 0 \\ 2r - 2t = 0. \end{cases}$$

Its augmented matrix and reduced row echelon form follow.

$$\left[\begin{array}{ccc|c} 1 & -1 & -1 & -1 \\ 1 & 1 & -1 & 1 \\ 2 & -1 & -3 & 0 \\ 2 & 0 & -2 & 0 \end{array} \right] \xrightarrow{\text{rref}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

So $r=-1$, $s=1$, $t=-1$ give the intersection point. We can check that it is the

same for the line and the plane: the intersection point is $\vec{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + (-1) \begin{bmatrix} 1 \\ 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ -1 \\ -2 \end{bmatrix}$

$$\text{and } \vec{x} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} + 1 \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + (-1) \begin{bmatrix} 1 \\ 1 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ -1 \\ -2 \end{bmatrix}.$$

Example: Find a plane perpendicular to $\vec{x} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \\ 3 \\ 2 \end{bmatrix}$ going through $\begin{bmatrix} 0 \\ -1 \\ -1 \\ 2 \end{bmatrix}$.

(i) In equation form.

(ii) In parametric form.

(i) We are given two directions to which we must be perpendicular, the vectors

$$\vec{v} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} \text{ and } \vec{w} = \begin{bmatrix} 1 \\ 1 \\ 3 \\ 2 \end{bmatrix}, \text{ as well as a point } \vec{q} = \begin{bmatrix} 0 \\ -1 \\ -1 \\ 2 \end{bmatrix} \text{ that we have to go}$$

through. An equation form is then:

$$\begin{cases} 0 = (\vec{x} - \vec{q}) \cdot \vec{v} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} - \begin{bmatrix} 0 \\ -1 \\ -1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} = x_1 - x_2 - 1 + x_3 + 1 \\ 0 = (\vec{x} - \vec{q}) \cdot \vec{w} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} - \begin{bmatrix} 0 \\ -1 \\ -1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 3 \\ 2 \end{bmatrix} = x_1 + x_2 + 1 + 3x_3 + 3 + 2x_4 - 4 \end{cases}$$

or equivalently:

ask

$$\begin{cases} x_1 - x_2 + x_3 = 0 \\ x_1 + x_2 + 3x_3 + 2x_4 = 0 \end{cases}$$

(ii) We solve the above system of equations to find a parametric form.

ask: is there other way? Yes, from above (x)

$$\left[\begin{array}{cccc|c} 1 & -1 & 1 & 0 & 0 \\ 1 & 1 & 3 & 2 & 0 \end{array} \right] \xrightarrow{\text{ref}} \left[\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \end{array} \right]$$

let x_1 be a parameter, leaving $x_4 = t$ and $x_3 = s$ as parameters, the last augmented matrix gives:

x_2 a parameter, and substitute x_3 into x_4 .

$$\begin{cases} x_1 + 2s + t = 0 \\ x_2 + s + t = 0 \end{cases} \quad \text{equivalently} \quad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2s - t \\ -s - t \\ s \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} -2 \\ -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

Example: Determine if any of the following vectors are parallel to the plane

$$\vec{x} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} -2 \\ -1 \\ 1 \\ 2 \end{bmatrix} + t \begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$$

(i) $\begin{bmatrix} 2 \\ -1 \\ 1 \\ 1 \end{bmatrix}$

(ii) $\begin{bmatrix} 3 \\ -3 \\ 1 \\ 1 \end{bmatrix}$

(iii) $\begin{bmatrix} -1 \\ 1 \\ 5 \\ 1 \end{bmatrix}$

Justify it without solving a system of linear equations.

Justify it using projections.

ask: how do we do this quick in 3D?

→ dot product with normal vector.

(i) The vector $\begin{bmatrix} 2 \\ -1 \\ 1 \\ 1 \end{bmatrix}$ is parallel to the plane if and only if it is a linear combination

of $\begin{bmatrix} -2 \\ -1 \\ 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$. We are solving the equation $\begin{bmatrix} 2 \\ -1 \\ 1 \\ 1 \end{bmatrix} = s \cdot \begin{bmatrix} -2 \\ -1 \\ 1 \\ 2 \end{bmatrix} + t \cdot \begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$

or equivalently the system of equations:

$$\begin{cases} -2s - t = 2 \\ -s - t = -1 \\ s + t = 1 \\ 2s + t = 1 \end{cases} \quad \text{with augmented matrix} \quad \left[\begin{array}{cc|c} -2 & -1 & 2 \\ -1 & -1 & -1 \\ 1 & 1 & 1 \\ 2 & 1 & 1 \end{array} \right] \xrightarrow{\text{rref}} \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{array} \right].$$

There are no solutions, so $\begin{bmatrix} 2 \\ -1 \\ 1 \\ 1 \end{bmatrix}$ is not a linear combination of $\begin{bmatrix} -2 \\ -1 \\ 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$,

so $\begin{bmatrix} 2 \\ -1 \\ 1 \\ 1 \end{bmatrix}$ is not parallel to the plane.

(ii) We can proceed as before, but we are not allowed to solve a system of

equations. Sometimes we can also use angles to determine linear dependence

or linear independence, and essentially we are just being asked if the vectors

$\begin{bmatrix} -2 \\ -1 \\ 1 \\ 2 \end{bmatrix}$, $\begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 3 \\ -3 \\ 1 \\ 1 \end{bmatrix}$ are linearly dependent or linearly independent. Note that the

last vector is orthogonal to the other two because:

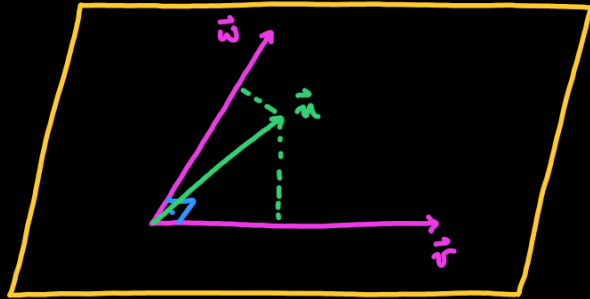
$$\begin{bmatrix} -2 \\ -1 \\ 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ -3 \\ 1 \\ 1 \end{bmatrix} = (-2) \cdot 3 + (-1) \cdot (-3) + 1 \cdot 1 + 2 \cdot 1 = 0$$

$$\begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ -3 \\ 1 \\ 1 \end{bmatrix} = (-1) \cdot 3 + (-1) \cdot (-3) + (-1) \cdot 1 + 1 \cdot 1 = 0$$

so $\begin{bmatrix} 3 \\ -3 \\ 1 \\ 1 \end{bmatrix}$ is not a linear combination of $\begin{bmatrix} -2 \\ -1 \\ 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$, so $\begin{bmatrix} 3 \\ -3 \\ 1 \\ 1 \end{bmatrix}$ is not

parallel to the plane.

(iii) The vector $\begin{bmatrix} -1 \\ 1 \\ 5 \\ 1 \end{bmatrix}$ is parallel to the plane if and only if it is equal to the sum of its projection onto two perpendicular vectors inside the plane.



We choose $\begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$ to be the first direction. We then decompose $\begin{bmatrix} -2 \\ -1 \\ 1 \\ 2 \end{bmatrix}$ into a component parallel to $\begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$ and a component orthogonal to $\begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$. The parallel component is:

$$\text{proj}_{\begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix}} \begin{bmatrix} -2 \\ -1 \\ 1 \\ 2 \end{bmatrix} = \left(\begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ -1 \\ 1 \\ 2 \end{bmatrix} \right) \cdot \frac{1}{\left\| \begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix} \right\|^2} \cdot \begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix} = \frac{2+1-1+2}{1+1+1+1} \cdot \begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix}.$$

The orthogonal component is:

$$\begin{bmatrix} -2 \\ -1 \\ 1 \\ 2 \end{bmatrix} - \text{proj}_{\begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix}} \begin{bmatrix} -2 \\ -1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \\ 1 \\ 2 \end{bmatrix} - \begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 2 \\ 1 \end{bmatrix}.$$

We now compute the projections:

$$\text{proj}_{\begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix}} \begin{bmatrix} -1 \\ 1 \\ 5 \\ 1 \end{bmatrix} = \left(\begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 1 \\ 5 \\ 1 \end{bmatrix} \right) \cdot \frac{1}{\left\| \begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix} \right\|^2} \cdot \begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix} = \frac{1-1-5+1}{1+1+1+1} \cdot \begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix} = - \begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$$

$$\text{proj}_{\begin{bmatrix} -1 \\ 0 \\ 2 \\ 1 \end{bmatrix}} \begin{bmatrix} -1 \\ 1 \\ 5 \\ 1 \end{bmatrix} = \left(\begin{bmatrix} -1 \\ 0 \\ 2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 1 \\ 5 \\ 1 \end{bmatrix} \right) \cdot \frac{1}{\left\| \begin{bmatrix} -1 \\ 0 \\ 2 \\ 1 \end{bmatrix} \right\|^2} \cdot \begin{bmatrix} -1 \\ 0 \\ 2 \\ 1 \end{bmatrix} = \frac{1+0+10+1}{1+0+4+1} \cdot \begin{bmatrix} -1 \\ 0 \\ 2 \\ 1 \end{bmatrix} = 2 \cdot \begin{bmatrix} -1 \\ 0 \\ 2 \\ 1 \end{bmatrix}$$

and we add them up:

$$\text{proj}_{\begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix}} \begin{bmatrix} -1 \\ 1 \\ 5 \\ 1 \end{bmatrix} + \text{proj}_{\begin{bmatrix} -1 \\ 0 \\ 2 \\ 1 \end{bmatrix}} \begin{bmatrix} -1 \\ 1 \\ 5 \\ 1 \end{bmatrix} = - \begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix} + 2 \cdot \begin{bmatrix} -1 \\ 0 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 5 \\ 1 \end{bmatrix}.$$

Since both projections are in the plane determined by $\begin{bmatrix} -2 \\ -1 \\ 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$, and $\begin{bmatrix} -1 \\ 1 \\ 5 \\ 1 \end{bmatrix}$ is a linear combination of those projections, the vector $\begin{bmatrix} -1 \\ 1 \\ 5 \\ 1 \end{bmatrix}$ is parallel to the plane.

Example: Find the smallest distance between the lines $\vec{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + s \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and

$$\vec{x} = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix} + t \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}.$$

These lines do not intersect because $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$ are parallel. The vector

$\vec{v} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$ is perpendicular to both of them, we can find it by solving the

equation $0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \cdot \vec{v}$. A point in the first line is $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, a point in the second line is $\begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}$, so the vector $\vec{w} = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$ points from the

first line to the second line. The length of the projection:

$$\text{proj}_{\vec{v}} \vec{w} = (\vec{v} \cdot \vec{w}) \frac{\vec{v}}{\|\vec{v}\|^2} = \frac{2+1}{1+1} \cdot \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = \frac{3}{2} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix},$$

namely:

$$\left\| \frac{3}{2} \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\| = \sqrt{\frac{9}{4} \cdot 1 + \frac{9}{4} \cdot 1} = \frac{3}{2} \cdot \sqrt{2},$$

is the distance between the lines.

Example: Find the smallest distance between the lines $\vec{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + s \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and

$$\vec{x} = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} + t \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}.$$

As before, the idea is to find a vector perpendicular to both lines, find a vector from one line to the other, and compute the length of the

projection. The cross product:

$$\vec{v} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \times \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} = \det \begin{bmatrix} \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \\ 1 & 1 & 1 \\ 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

is perpendicular to both lines, the vector $\vec{w} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$ points from the

first line to the second, and the length of the projection:

$$\text{proj}_{\vec{v}} \vec{w} = (\vec{v} \cdot \vec{w}) \frac{\vec{v}}{\|\vec{v}\|^2} = \frac{-2+0-1}{1+0+1} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \frac{-3}{2} \cdot \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix},$$

namely:

$$\left\| \frac{-3}{2} \cdot \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\| = \frac{3}{2} \cdot \sqrt{1+0+1} = \frac{3\sqrt{2}}{2}, \text{ is the distance between the lines.}$$

Example: Find the smallest distance between the lines $\vec{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + s \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$ and

$$\vec{x} = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 2 \end{bmatrix} + t \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix}.$$

The idea is still the same, find a vector perpendicular to both lines, find a vector from one line to the other, and compute the length of the projection. However, since now there is a two dimensional plane parallel to each line, we need to be more careful when finding a vector perpendicular to both of them. For this, it is useful to first find the general vector pointing from the first line to the second line, and then impose that this vector be perpendicular to both lines. The general vector pointing from the first line to the second is:

$$\vec{w} = \left(\begin{bmatrix} -1 \\ 0 \\ 1 \\ 2 \end{bmatrix} + t \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix} \right) - \left(\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + s \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} -2 \\ -1 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix} + s \begin{bmatrix} -1 \\ 1 \\ -1 \\ 1 \end{bmatrix}.$$

Imposing perpendicularity to the lines gives:

$$\begin{cases} 0 = \vec{w} \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} = -2 + 1 + 0 - 1 + 2t - t + t - 2t - s - s - s - s = -2 - 4s \\ 0 = \vec{w} \cdot \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix} = -4 - 1 + 0 + 2 + 4t + t + t + 4t - 2s + s - s + 2s = -3 + 10t \end{cases}$$

whose solution is $s = -\frac{1}{2}$ and $t = \frac{3}{10}$. Substituting these into $\vec{w}$ gives:

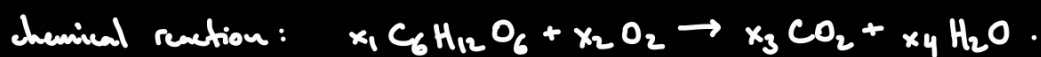
$$\begin{bmatrix} -2 \\ -1 \\ 0 \\ 1 \end{bmatrix} + \frac{3}{10} \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} -1 \\ 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -9/10 \\ -6/5 \\ 4/5 \\ 11/10 \end{bmatrix}$$

whose length:

$$\left\| \begin{bmatrix} -9/10 \\ -6/5 \\ 4/5 \\ 11/10 \end{bmatrix} \right\| = \frac{1}{10} \cdot \sqrt{81 + 144 + 64 + 121} = \frac{\sqrt{410}}{10} = \sqrt{\frac{41}{10}}$$

is the distance between the two lines.

Example: Find the smallest natural numbers x_1, x_2, x_3, x_4 that balance the following



We want to have the same number of each element before and after the reaction.

$$\begin{array}{ll} \text{C:} & 6x_1 = x_3 \\ \text{H:} & 12x_1 = 2x_4 \\ \text{O:} & 6x_1 + 2x_2 = 2x_3 + x_4 \end{array} \quad \text{namely} \quad \begin{cases} 6x_1 - x_3 = 0 \\ 12x_1 - 2x_4 = 0 \\ 6x_1 + 2x_2 - 2x_3 - x_4 = 0 \end{cases}$$

The augmented matrix and its ref are the following.

$$\left[\begin{array}{cccc|c} 6 & 0 & -3 & 0 & 0 \\ 12 & 0 & 0 & -2 & 0 \\ 6 & 2 & -2 & -1 & 0 \end{array} \right] \xrightarrow{\text{ref}} \left[\begin{array}{cccc|c} 6 & 0 & -1 & 0 & 0 \\ 0 & 2 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{array} \right]$$

Then:

$$\begin{cases} 6x_1 - x_4 = 0 \\ 2x_2 - x_3 - x_4 = 0 \\ x_3 - x_4 = 0 \end{cases} \quad \text{so setting } x_4 \text{ as a parameter gives} \quad \begin{cases} x_1 = \frac{1}{6} x_4 \\ x_2 = x_4 \\ x_3 = x_4 \\ x_4 = x_4. \end{cases}$$

The smallest natural solutions occur for $x_4 = 6$. Then:



Example: distance from point to a plane.